STA642 - Probability Distributions (Quiz 3)

Question # 1 of 5 (Start time: 07:44:13 PM, 27 August 2022)

If

$$f_X(x) = \begin{cases} \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

then on the value of $\alpha = 1$, which distribution we obtained?

Select the correct option

○	Poisson
○	None
●	Exponential
○	Normal

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STA642 - Probability Distributions (Quiz 3) **Quiz Start Time: 07:44 PM, 27 August 2022**

Question # 2 of 5 (Start time: 07:45:00 PM, 27 August 2022) **Total Marks:**

If we increase the value of $\alpha = 1 to 5$ with the $\lambda = 1/2$ then the shape of the distribution turns in to --------- distribution.

Select the correct option

Reload Math Equations

- (●) Normal
- ( ) flatter than before
- ( ) negatively skewed
- ( ) positively skewed

Click to Save Answer & Move to Next Question

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STA642 - Probability Distributions (Quiz 3) **Quiz Start Time: 07:44 PM, 27 August 2022**

Question # 3 of 5 (Start time: 07:45:46 PM, 27 August 2022) **Total Marks:**

The mean of exponential distribution is given as _____

Select the correct option

Reload Math Equations

- $(\mu)^2$
- μ
- $\frac{1}{\mu}$
- $\frac{1}{\mu^2}$

Click to Save Answer & Move to Next Question

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STA642 - Probability Distributions (Quiz 3) **Quiz Start Time: 07:44 PM, 27 August 2022**

Question # 4 of 5 (Start time: 07:46:01 PM, 27 August 2022) **Total Marks:** 1

The pdf of poisson distribution is

Select the correct option

Reload Math Equations

- $\frac{e^{m}m^{x}}{x!}$
- $\frac{x!}{m^{x}e^{-m}}$
- $\frac{e^{-m}m^{x}}{x!}$
- $\frac{e^{-m}x!}{m^{x}}$

Click to Save Answer & Move to Next Question

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